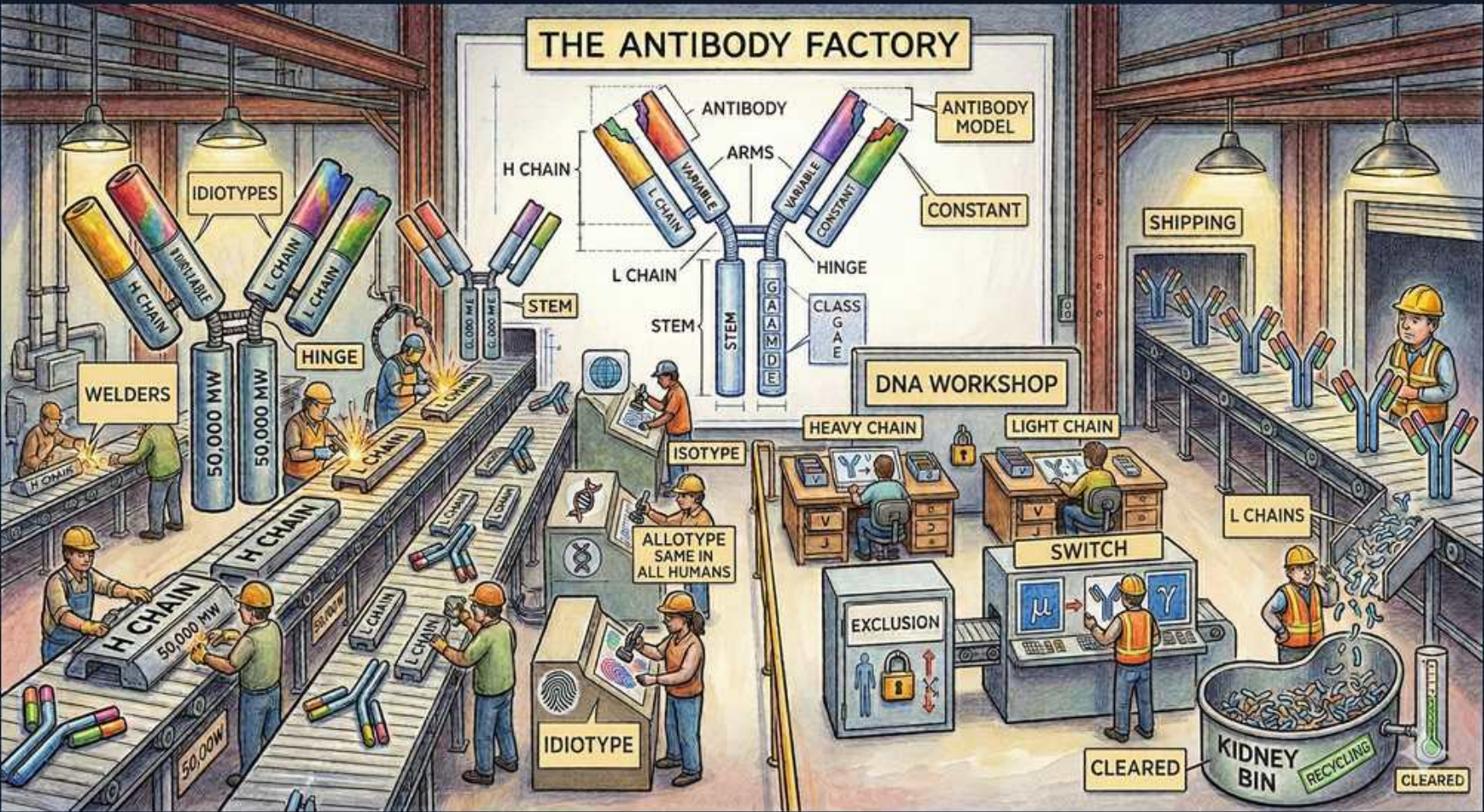


Immunoglobulin Structure — The Antibody Factory



1. Antibody / Y structure

WHAT YOU SEE

Central blueprint of a Y-shaped antibody with arms and stem labeled.

WHAT IT ENCODES

An immunoglobulin is a Y-shaped glycoprotein made by plasma cells: two identical heavy chains and two identical light chains linked by disulfide bonds. The arms are the antigen-binding Fab regions; the stem is the Fc effector region.

BOARD TRAP

Each antibody monomer has exactly TWO antigen-binding sites (bivalent) — except secreted IgM (pentamer, 10 sites) and dimeric secretory IgA.

2. Heavy chains (50,000 MW)

WHAT YOU SEE

H CHAIN welders, 50,000 MW.

WHAT IT ENCODES

Heavy chains (~50 kDa) determine the isotype: μ (IgM), γ (IgG), α (IgA), δ (IgD), ϵ (IgE). Class switching changes the heavy-chain constant region while keeping antigen specificity.

BOARD TRAP

The heavy-chain constant region defines the CLASS/isotype and Fc function; the heavy-chain variable region (with light chain) defines specificity — different jobs.

3. Light chains (kappa/lambda)

WHAT YOU SEE

L CHAIN line, 25,000 MW, kappa/lambda.

WHAT IT ENCODES

Light chains (~25 kDa) are either kappa or lambda — never both in one antibody (isotypic exclusion). Normal serum $\kappa:\lambda$ ratio ~2:1; free light chains are filtered and reabsorbed by the kidney.

BOARD TRAP

A skewed serum free light-chain ratio (or monoclonal light chain) signals a clonal plasma-cell process (myeloma/MGUS/AL amyloid), not a polyclonal reactive response.

4. Variable region / idiotype

WHAT YOU SEE

IDIOTYPES at the antibody tips; idiotype label at bottom center.

WHAT IT ENCODES

The variable regions (VH+VL) form the antigen-binding site; the unique set of hypervariable (CDR) sequences defining one antibody's specificity is its IDIOTYPE. Idiotypic diversity is generated by V(D)J recombination.

BOARD TRAP

Idiotype = the antigen-binding specificity (varies clone to clone); do not confuse with isotype (class) or allotype (allelic variant).

5. Constant region / Fc

WHAT YOU SEE

CONSTANT label on the antibody stem.

WHAT IT ENCODES

The constant regions form the Fc stem, which mediates effector functions: complement activation (classical pathway), opsonization, and binding to Fc receptors on phagocytes/NK cells (ADCC).

BOARD TRAP

The Fab binds antigen; the Fc carries out effector function (complement/Fc-receptor binding) — therapeutic 'naked' antibodies often kill via Fc-mediated ADCC/CDC.

6. Hinge region

WHAT YOU SEE

HINGE label between Fab arms and stem.

WHAT IT ENCODES

The flexible hinge between Fab arms and Fc lets the arms swing to bind antigens at variable distances and is the site cleaved by proteases (papain $\rightarrow$ 2 Fab + Fc; pepsin $\rightarrow$ F(ab')₂ + degraded Fc).

BOARD TRAP

Papain cuts ABOVE the hinge giving two monovalent Fab + Fc; pepsin cuts BELOW giving one bivalent F(ab')₂ — a classic immunology cleavage question.

7. Isotype

WHAT YOU SEE

ISOTYPE workstation.

WHAT IT ENCODES

Isotype = antibody class (IgG, IgM, IgA, IgD, IgE), determined by the heavy-chain constant region and shared by all healthy humans. IgG predominates in serum and crosses the placenta; IgM is the first/largest (pentamer).

BOARD TRAP

Isotype is shared across all people; allotype varies between individuals (allelic); idiotype varies between clones — three different concepts board exams love to swap.

8. Allotype (same in all humans of an allele)

WHAT YOU SEE

ALLOTYPE workstation: 'same in all humans'.

WHAT IT ENCODES

Allotypes are allelic (polymorphic) variants of constant regions inherited individually — they differ between people, not between antibody classes or clones.

BOARD TRAP

The visual cue can mislead: allotype is the INTER-individual (allelic) variation, distinct from isotype (class, shared by all) and idiotype (clonal specificity).

9. Class switch recombination

WHAT YOU SEE

SWITCH station in the DNA workshop.

WHAT IT ENCODES

Class (isotype) switching is mediated by AID in germinal centers, changing the heavy-chain constant region (IgM→IgG/IgA/IgE) while preserving antigen specificity. It is T-cell/CD40L dependent.

BOARD TRAP

Hyper-IgM syndrome (CD40L or AID defect) blocks class switching — high IgM, low IgG/IgA/IgE, with recurrent infections and PJP risk.

10. Isotypic exclusion

WHAT YOU SEE

EXCLUSION lock station.

WHAT IT ENCODES

Allelic and isotypic exclusion ensure each B cell/plasma cell makes a single heavy chain allele and a single light-chain type (κ OR λ), producing one specificity (monospecificity).

BOARD TRAP

This is why a monoclonal gammopathy shows a single restricted light chain — clonal plasma cells obey exclusion, so the spike is κ-restricted OR λ-restricted, never both.

11. Free light chains → kidney

WHAT YOU SEE

KIDNEY BIN conveyor with CLEARED light chains, L CHAINS shipping.

WHAT IT ENCODES

Excess free light chains are small enough to be filtered by the glomerulus and reabsorbed/catabolized in the proximal tubule. Overproduction (myeloma) overwhelms this, causing cast nephropathy (Bence Jones proteinuria) and AL amyloid.

BOARD TRAP

Urine dipstick detects albumin, NOT light chains — Bence Jones (light-chain) proteinuria needs urine protein electrophoresis/immunofixation, so a negative dipstick does not exclude myeloma.

12. Class/isotype determination

WHAT YOU SEE

CLASS/IgG gate in the DNA workshop, heavy chain to light chain assembly.

WHAT IT ENCODES

Heavy + light chain assembly in the plasma cell yields the complete immunoglobulin; the heavy-chain constant gene segment chosen sets the class and thus serum half-life, complement-fixing ability, and tissue distribution.

BOARD TRAP

IgG has the longest serum half-life (~21 days, FcRn-recycled) and is the only isotype that crosses the placenta — key facts the 'class' choice encodes.
